

Deterministic Optimization

Further Development of Simplex Method

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Simplex Method Example

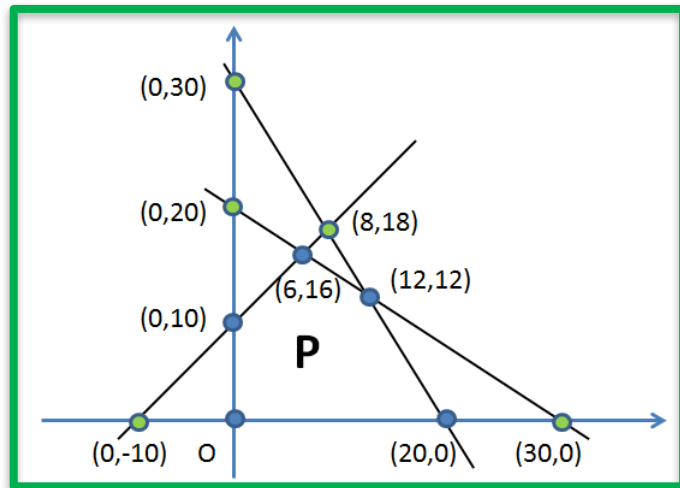
Further Development of Simplex Method

Learning Objectives for this lesson

- Put Simplex Method in action to solve an example problem

Problem Setup

$$\begin{array}{ll}
 \max & 2x_1 + 3x_2 \\
 \text{s.t.} & -x_1 + x_2 \leq 10 \\
 & 3x_1 + 2x_2 \leq 60 \\
 & 2x_1 + 3x_2 \leq 6 \\
 & x_1, x_2 \geq 0.
 \end{array}$$



$$\begin{array}{llllll}
 \min & -2x_1 & -3x_2 & & & \\
 \text{s.t.} & -x_1 & +x_2 & +x_3 & & = 10 \\
 & 3x_1 & +2x_2 & & +x_4 & = 60 \\
 & 2x_1 & +3x_2 & & & +x_5 = 60 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0.
 \end{array}$$

Iteration 1 (Step 1)

1. **Choose a starting BFS:** Let us select the basis $\mathbf{B} = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The corresponding basic solution is

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 10 \\ 60 \\ 60 \end{bmatrix}, \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and the cost coefficients associated with basic and nonbasic variables:

$$\mathbf{c}_B = \begin{bmatrix} c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{c}_N = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}.$$

Since $\mathbf{x}_B \geq \mathbf{0}$ (and of course $\mathbf{x}_N \geq \mathbf{0}$), the current basic solution is a basic feasible solution. So we are ready to start the simplex method.

Iteration 1 (Step 2)

2. Compute reduced costs for nonbasic variables:

$$\bar{c}_1 = c_1 - \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A}_1 = -2$$

$$\bar{c}_2 = c_2 - \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A}_2 = -3$$

Both $\bar{c}_1$ and $\bar{c}_2$ are negative. Therefore, the current BFS is not optimal, and both x_1 and x_2 are candidates to enter the basis, i.e. to increase to a positive value. Let us take x_2 to enter the basis, and keep x_1 zero.

Iteration 1 (Step 3)

3. **Compute feasible direction $\mathbf{d} = \begin{bmatrix} \mathbf{d}_B \\ \mathbf{d}_N \end{bmatrix}$:** Since we decide to increase x_2 and keep x_1 at zero, the nonbasic variable part of the feasible direction $\mathbf{d}_N$ is $\mathbf{d}_N = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and the basic variable part of the feasible direction $\mathbf{d}_B$ is

$$\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_2 = \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}.$$

Since some components of $\mathbf{d}_B$ is negative, we do not have an unbounded optimal solution, and we need to decide how far to go along this direction while still remaining feasible.

Iteration 1 (Step 4)

4. **Min-ratio test:** By going along the above calculated direction, we are going from the initial BFS $\mathbf{x}$ to a new point $\mathbf{x} + \theta \mathbf{d}$. Let us write it out componentwise:

$$\mathbf{x} + \theta \mathbf{d} = \begin{bmatrix} \mathbf{x}_B + \theta \mathbf{d}_B \\ \mathbf{x}_N + \theta \mathbf{d}_N \end{bmatrix} = \begin{bmatrix} x_{B(1)} + \theta d_{B(1)} \\ x_{B(2)} + \theta d_{B(2)} \\ x_{B(3)} + \theta d_{B(3)} \\ x_1 + \theta \cdot 0 \\ x_2 + \theta \cdot 1 \end{bmatrix} = \begin{bmatrix} x_3 + \theta d_3 \\ x_4 + \theta d_4 \\ x_5 + \theta d_5 \\ x_1 \\ x_2 + \theta \end{bmatrix} = \begin{bmatrix} 10 + \theta \cdot (-1) \\ 60 + \theta \cdot (-2) \\ 60 + \theta \cdot (-3) \\ 0 \\ \theta \end{bmatrix} = \begin{bmatrix} 10 - \theta \\ 60 - 2\theta \\ 60 - 3\theta \\ 0 \\ \theta \end{bmatrix}$$

To decide the largest θ so that $\mathbf{x} + \theta \mathbf{d} \geq \mathbf{0}$, we need to do the min-ratio test:

$$\theta^* = \min_{\{i=1, \dots, m \mid d_{B(i)} < 0\}} \frac{x_{B(i)}}{-d_{B(i)}} = \min\left\{\frac{10}{1}, \frac{60}{2}, \frac{60}{3}\right\} = 10.$$

So $x_{B(1)} = x_3$ exits the basis.

Iteration 1 (Step 5)

5. **The new basis:** The new basis $\bar{\mathbf{B}} = [\mathbf{A}_2, \mathbf{A}_4, \mathbf{A}_5]$, which differs from the original basis only in one column: $\mathbf{A}_3$ is replaced by $\mathbf{A}_2$ i.e. x_3 exits the basis and x_2 enters the basis. The new basic variables and nonbasic variables are

$$\mathbf{x}_{\bar{\mathbf{B}}} = \begin{bmatrix} x_2 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 10 \\ 40 \\ 30 \end{bmatrix}, \quad \mathbf{x}_{\bar{\mathbf{N}}} = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

We are ready for a new iteration of the simplex method.

Summary

- We learned:
 - Detailed steps of simplex method
 - You should be able to carry out the remaining iterations until the simplex method finds the optimum.